Telematics and Data Collection

A practical guide to what the system checks, what it collects, what each measurement means, where data goes, and what the numbers do - for rental telematics and the separate RC Research collector.

Status	Implementation is present in code; physical Android/iOS validation remains pending for sensors, GPS delivery, background behavior, and geofence crossings.
Coverage	Self Drive (Customer operator), With Assigned Driver (currently assigned Driver operator), and the isolated Admin RC Research collector.
One key distinction	Rental telematics keeps bounded calibration summaries and a single encrypted latest GPS point. RC Research separately preserves reviewed raw motion samples for future research.

System at a glance

Laravel decides who may act, validates evidence, controls setup and release, stores the rental state, and evaluates geofence transitions. The mobile app presents the workflow and uses native phone APIs through service modules. Staff controls handover and the existing **Start Rental** action releases the vehicle. The phone operator never releases a rental.

The capstone goal is AI driving-event classification, but no classifier or driver scoring is implemented. Rental GPS and setup calibration do not create an AI dataset. The separate RC Research collector is the only current workflow that saves raw motion samples, and it is a local/testing research pilot.

What is in scope

Phone readiness, consented supervised calibration, explicit post-release location sharing, frozen geofence monitoring, staff map/activity views, and a separate RC motion-study workflow. Video, audio, dashcam uploads, raw rental accelerometer/gyroscope streams, automated fault findings, and automatic penalties are outside the scope.

1. Rental lifecycle: from setup to a released trip

The operator is the owning Customer for Self Drive and the current assigned Driver for With Assigned Driver. Staff finalizes the Outgoing Inspection and starts handover. Setup checks can be submitted only for the authorized operator, a confirmed booking, and a staff-started handover. The renter cannot start handover or release.

Stage	What happens	Data boundary
Staff preparation	Finalize signed Outgoing Inspection, then Start Handover.	No telemetry starts.
Device readiness	Six checks: device capabilities; motion permission; foreground location permission; background location permission/task readiness; one fresh GPS quality check; explicit secure phone-mount confirmation.	One setup GPS fix is evaluated and then discarded. No sensor stream, trip, background update, or zone check.
Calibration	Five ordered, supervised foreground captures: Stationary, Forward, Braking, Left turn, Right turn. Laravel evaluates each typed summary.	Only bounded statistics/reference summaries are submitted and stored; raw motion arrays are not uploaded or retained.
Ready for release	All eleven latest required steps pass in standard mode. Reassignment requires fresh setup by the new Driver.	Attempt history is append-only; latest effective result determines readiness.
Staff release	Authorized Start Rental moves the booking to ongoing and freezes all active default and custom geofences.	At most one rental trip can exist. This is the only release path.
Active sharing	After release, operator explicitly agrees to foreground-only or foreground-and-background GPS sharing. A setup-time preparation can trigger one start after a fresh release check only while the same prepared screen/runtime is still active.	Latest location only for this sharing flow; no route history or raw motion. Stop and interruption require explicit recovery.

Server timing is authoritative. The standard handover interval is the reserved rental interval. A bounded local/testing same-start-date mode exists, but does not fabricate evidence; the separate broad handover test mode is also local/testing only and has a documented Self Drive mount-only readiness exception. Neither changes the production release rule.

The system has one pre-release setup session and at most one post-release trip. Setup/calibration cannot create a trip. Pausing/resuming location sharing remains inside the same trip.

2. Device readiness and GPS checks

Readiness is a checklist, not telemetry collection. The operator sees a disclosure before checks. Checks run in order and stop at the first failure in standard mode; rerunning starts fresh checks. OS permissions already granted do not prove a new grant or validate sensor behavior. Phone mount is a human confirmation, not automatic holder detection.

Check / number	Meaning
Accelerometer + gyroscope availability	The phone says motion capability exists; no motion subscription is started.
Motion permission	Required for later calibration captures. A denial is recoverable through device Settings.
Foreground location permission	Needed for the one setup GPS fix and later rental sharing.
Background location permission + inert task readiness	Permission and task availability only. Setup does not register location updates. Android may use Settings; iOS background permission must be prepared.
Fresh GPS fix	Latitude -90..90, longitude -180..180; services enabled; captured no more than 60 seconds ago and no more than 10 seconds in the future; horizontal accuracy at most 50 m.
Phone mount confirmation	Operator explicitly confirms the phone is securely mounted. A false or dismissed answer does not pass.

A setup fix checks location quality only. Geofence membership does not affect readiness. Exact coordinates are not retained in step evidence or returned to the mobile handover read.

All six checks and all five calibration exercises are required for standard release. These device checks alone leave setup in progress. Local/testing exceptions are not physical evidence and are not standard production behavior.

3. Rental calibration: what the motion numbers mean

Calibration is an explicit, supervised foreground measurement. Android/iOS DeviceMotion values are platform-processed, not unprocessed hardware data. The capture service computes bounded statistics in memory; it does not upload or store raw sample arrays. Motion access is needed, but calibration does not ask for GPS or start the rental trip.

Measure	Definition / unit
Linear acceleration XYZ	Acceleration with gravity excluded; m/s^2. Axes are phone device coordinates.
Gravity XYZ	Derived as acceleration-including-gravity minus linear acceleration; m/s^2. Stationary establishes the down reference.
Rotation rate XYZ	Platform-normalized device-axis angular rate; degrees/second (deg/s). Axis order differs by platform and is normalized in code. Physical signs still need Android/iOS validation.
RMS	Root mean square of 3D vector magnitudes. Acceleration RMS is m/s^2; rotation RMS is deg/s. RMS includes sustained mean.
STD	Composite population variation of XYZ samples around their mean; not sample standard error and not a custom noise filter.
Forward window	A contiguous set of qualifying horizontal acceleration samples. Mean vector direction becomes the learned forward reference only after server acceptance.
Coherence	0 to 1 agreement of contributing window vector directions. It is not probability or model confidence.
Yaw left / right	Gyroscope projection onto the gravity reference integrated over valid time intervals; degrees of short-window rotation, not absolute heading.
Tilt	Change in gravity direction from the established reference; degrees. Detects some mount movement, but not every phone rotation.

Statistics use equally weighted accepted samples. There is no custom continuous sensor-fusion estimator or bias-correction pass. The system combines platform acceleration, gravity and rotation geometrically for the bounded calibration tasks.

Exercise order and outcome

Stationary measures the complete parked interval and supplies a noise baseline. Forward looks for a consistent direction window above the server-derived acceleration minimum. Braking checks the learned longitudinal direction with the opposite sign. Left/right turns use gravity-projected yaw. Laravel, not the client, decides pass/fail. A capture timeout does not pass missing evidence.

4. Rental calibration: current numeric settings

All values below are configurable development defaults or provisional engineering limits. They are not automotive standards, validated safe-driving thresholds, or AI labels.

Value	Where it applies	Meaning
50 ms / 20 Hz	Calibration request	Requested DeviceMotion update interval; actual delivery can vary.
200 ms	Fresh-stream startup	Requires advancing sensor timestamps that track callback arrival before recording begins. Startup failure after 3 seconds creates no attempt.
50 ms	Per-channel timestamp alignment	Acceleration, gravity and rotation timestamps must be close (100 ms tolerance).
1,000 ms	Minimum movement capture	Movement evidence must include at least one second; Stationary has its full configured duration.
6 seconds	Stationary	Full parked interval; no trimming noisy sub-windows.
12 / 12 / 15 / 15 seconds	Movement maximums	Forward, Braking, Left turn, Right turn respectively; bounded timeouts. RC demo profile uses 3 seconds for movement steps.
8 samples/second	Shared quality minimum	At least this average sample rate.
85%	Shared coverage minimum	Sensor timestamp span must cover at least this fraction of capture duration.
500 ms maximum gap	Shared summary gate	A larger maximum observed gap fails the shared check. Window continuity is stricter: over 250 ms breaks a Forward candidate.
8..12 m/s^2	Gravity plausibility	Mean gravity magnitude range.
10 degrees	Tilt limit	Maximum tilt relative to accepted reference.
0.35 m/s^2	Stationary linear RMS ceiling	Full parked capture must remain at or below this value.
6 deg/s	Stationary/Forward rotation peak ceiling	Peak rotation-rate limit.
5 degrees	Stationary/Forward total yaw ceiling	Sum of left and right yaw must stay at or below this value.
max(0.5 m/s^2, 3 x parked horizontal RMS)	Forward threshold	A floor plus three times the accepted parked horizontal RMS; engineering margin, not three-sigma confidence.
500 ms, 5 samples, 0.9	Forward window	At least 500 ms, at least 5 samples, coherence at least 0.9. Client target is 750 ms; final interval can extend to 1 second.
-0.5 m/s^2	Braking	Minimum longitudinal acceleration in expected direction.
20 degrees / 10 degrees	Turns	At least 20 degrees in expected direction and at most 10 degrees in the opposite direction.
60 s / 10 s	Submission clock window	Capture end is at most 60 seconds old and no more than 10 seconds in the future.
8 x duration to 30 x duration + 5	Sample count plausibility bounds	Server requires a duration-based minimum sample count and caps excessive counts; these checks are separate from the minimum 8 samples/second.

5. Rental GPS collection: what is sent and what is saved

Location sharing applies only after staff release. The operator chooses foreground-only or foreground-and-background in the disclosure. The app never silently falls back to a different mode. Opening a rental, completing setup, login, granting permission, or returning to the app does not by itself start collection.

Item	Current behavior
Location fields	Latitude, longitude, horizontal accuracy, captured_at. Speed, heading, motion values and client-claimed zones are prohibited.
Foreground request	Android asks for High accuracy at a requested 5-second interval and zero distance threshold; iOS controls delivery timing. App caps uploads at one per 5 seconds.
Freshness	Only fixes captured after the explicit sharing action are eligible. Cached pre-consent fixes are discarded. No initial usable fix within 30 seconds stops sharing. Upload request timeout is 10 seconds.
GPS quality bounds	At most 50 m horizontal accuracy; captured no more than 60 seconds ago or 10 seconds in the future.
Buffer and offline	One pending latest point can be replaced by newer data. There is no disk cache or offline route buffer. Network failure stops the watch and discards pending data.
Trip behavior	The first accepted fix starts or reuses the single trip. Resume reuses that trip. Sharing Stop pauses; physical receipt or trip end clears the point and closes the session.
Server authorization	Each start/update rechecks current operator, ongoing booking, release, receipt/trip state, token and capture freshness. Each session token is 64 characters, kept in app memory; only its SHA-256 hash is retained server-side.
Saved location	One Laravel-encrypted latest_location value with latitude, longitude and accuracy, plus a timestamp. It is not a route. Older/duplicate captured seconds cannot overwrite the latest point.
Retention	Latest point expires after 24 hours from last update. Hourly scheduler purge makes the disclosed upper bound 25 hours, only if Laravel scheduling runs.

Background mode

Android needs foreground and background location permission and uses a visible foreground-service notification. iOS requires Always access and shows the system location indicator. The native task uses a requested 5-second interval and 5-second deferred interval; OS power limits can change actual timing. Only the newest candidate in a native batch is considered. Foreground-only mode pauses when the app leaves the foreground.

The session and consent token remain in memory. Closing/force-stopping the app, runtime loss or reload stops collection. There is no cold-start or automatic restart. The operator must reopen the rental and agree again.

6. Geofence events and in-app alerts

Admin manages default circle/polygon zones in the web library and optional custom zones on a pending/confirmed booking. Staff can view but not manage zones. Active defaults automatically apply to each supported rental. At Start Rental, Laravel copies active defaults and freezes all zones plus entry/exit preferences. Later edits affect future rentals only. Zones do not gate setup, handover, release or GPS collection.

Laravel evaluates each frozen zone independently on accepted latest-GPS fixes. The first reliable location sets a baseline and is not an entry/exit event. Confirmed transitions create server-owned events. The customer receives an in-app alert when that direction is enabled; on With Assigned Driver, the current assigned Driver may also receive it. No coordinates or geometry are copied into event/notification data.

Number / rule	Interpretation
50 m	Maximum accepted horizontal accuracy for geofence input.
60 seconds	Maximum age for an accepted location fix. Future clock skew allowance is 10 seconds.
accuracy + 10 m	Boundary uncertainty band. A fix this close to a zone edge is uncertain; the last confirmed membership is retained and transition candidates reset.
2 observations / 10 seconds	A membership change needs at least two reliable observations spread over at least 10 seconds. This smooths noisy GPS; it does not promise exact crossing time.
3 points minimum	Minimum polygon vertices. No application-level maximum; large polygons increase request, storage and compute costs.
30 crossings	Booking details show the latest 30 confirmed crossings.
15 seconds	Mobile Activity refresh polling while focused; there is no phone push, email, or global staff notification bell.

Overlapping zones are independent. A resumed sharing session resets membership/candidate state; its first reliable fix is a new baseline, so crossings during the pause are not inferred.

7. Fleet Monitoring and what its status means

The web Fleet Monitoring module lists ongoing rentals, including overdue rentals and those without a trip. Authorized map reads display only latest points. Private Reverb messages are empty invalidation notices; the browser re-reads data through the authorized API.

Status	Meaning
Not started	Supported rental has no trip.
Awaiting location	Trip started but no saved location timestamp yet.
Recent location	Saved timestamp is within the 120-second display freshness cutoff.
No recent updates	Saved timestamp is older than that cutoff.
Paused	Saved trip status is paused, regardless of point age.
Ended	Trip ended while booking remains ongoing.
Unavailable	Setup has not been released.

The 120-second status window is for display/filtering. It is not a GPS sampling rate or a diagnosis that the phone, permission, GPS, network or vehicle has failed. A saved "started" status alone does not prove the app is connected or collecting.

Map refresh numbers

The visible tab performs a 30-second reconciliation read. Reverb notices are coalesced for 750 ms; subscription errors retry after 10 seconds; reads time out after 10 seconds and do not overlap. Visibility restoration, network restoration and manual refresh also reconcile missed notices.

8. Separate RC Research raw-motion collection

RC Research is a protected Admin workflow in local/testing RC mode. It is separate from customer/driver rentals and from rental calibration. It captures labeled smartphone motion trials on an RC car testbed. Consent is explicit. Start and Stop are manual; no countdown, automatic timer, background recording or auto-save exists. Leaving the foreground cancels active capture.

Saved sample field	Meaning / unit
elapsed_ms	Callback arrival time relative to recording start, milliseconds.
acceleration_timestamp_s / gravity_timestamp_s / rotation_timestamp_s	Native channel timestamps in seconds. Preserved separately.
linear_x/y/z	Platform-processed acceleration excluding gravity, m/s^2.
including_gravity_x/y/z	Platform-processed acceleration including gravity, m/s^2.
rotation_x/y/z	Rotation-rate axes normalized to device XYZ, degrees/second.
orientation	Screen-orientation value; not attitude or vehicle direction.

Unlike rental calibration, RC Research preserves the sample-level values in a reviewed recording. The first future model is planned to use derived numeric window features and confirmed observer labels. It is not implemented yet. DeviceMotion is platform-processed, not raw hardware output.

Current RC pilot numbers

Number	Meaning
20 ms / 50 Hz	Requested interval/rate; actual rate can vary.
200 ms	Warmup qualification before recording clock starts.
100 ms	Maximum difference allowed between channel timestamps; values beyond are rejected.
250 ms	Warmup gap ceiling and summary gap warning threshold.
3 seconds	Fresh stream startup timeout. No completed trial is produced.
1 second + 8 samples	Minimum capture.
8 Hz / 85% / 250 ms	Quality flags for low mean rate, limited coverage, and sample gaps, respectively. These do not assign normal/harsh labels.
10,000 samples / 4,000,000 JSON bytes	Maximum in-memory samples/payload. Reaching either bound fails the trial; partial data is not saved.
24 hours	Defensive maximum wall-clock duration check; not a recommended trial duration.
500 characters	Maximum reviewer note length.

9. RC Research labels, storage and retention

Review item	What it means
Movement	Stationary, steady movement, acceleration, braking, left turn, or right turn. Record observed movement, even when it differs from intended movement.
Intensity	Normal or harsh for acceleration/braking/turns; not applicable for stationary/steady.
Label state	Confirmed or uncertain. Ambiguous trials should remain uncertain and be excluded from the first supervised set.
Quality	Usable, interrupted, or mount disturbed. Mount disturbance is a quality flag, not a maneuver class.
Event interval	Capture-relative start/end in milliseconds. The event may occupy only part of the recording.
Note	Optional review note up to 500 characters.

Samples stay in memory until explicit review/save. Laravel validates finite numbers, supported enums/protocol, order/timing, count/size, ownership, content and annotation intervals. It writes an encrypted immutable private file; samples are not stored in database JSON, mobile list responses or ordinary rental endpoints. The same trial retry ID/content is idempotent; changed content or a deleted ID returns conflict.

Saved RC trials remain until the owning Admin explicitly deletes them; there is no age-based purge. Deletion removes ciphertext and clears notes/summaries but leaves a minimal tombstone to block interrupted replay. Session metadata remains. Unsent raw data is memory-only and is lost when the app closes. External backups follow operator policy.

Future collection/model work

The planned offline workflow is quality validation, timestamp/axis checks, feature generation, then a supervised baseline such as Random Forest using whole-session splits. The proposal uses 2-second windows with a 1-second stride, but that is not a fixed requirement. Training/test separation must be by whole sessions to avoid leakage. Do not treat thresholds as labels, model output as ground truth, or an RC-only result as full-size vehicle validation.

10. Privacy, evidence status and where to read more

Data boundary summary

Data	Rental flow	RC Research flow
Setup GPS	One fresh fix evaluated, then discarded.	Not collected.
Calibration motion	Bounded summaries only; no raw array.	Not applicable.
Released-rental GPS	Single encrypted latest point; no route-history points for this consented latest-only path.	Not collected.
Raw motion samples	Not collected during rental.	Explicitly captured, reviewed, uploaded and encrypted in private storage.
Events	Significant lifecycle/geofence occurrences; no raw coordinates in event or alert payload.	Trial labels/intervals/quality/notes stay separate from sample readings.
AI output	No driving-event classifier, driver score, fault or penalty.	Model training is future work; no classifier yet.

Evidence status

The workflows and server rules are implemented in code. That is distinct from physical validation. Android/iOS sensor signs, permission dialogs, GPS cadence, background delivery, geofence crossings and full end-to-end phone-to-map behavior still need physical device checks. Numeric values described as pilot/provisional/configurable remain engineering defaults until validated with appropriate data.

Project references

Rental system: docs/telematics-architecture.md; docs/telematics-device-location.md; docs/telematics-calibration.md; docs/telematics-calibration-computation.md; docs/telematics-foreground-location.md; docs/telematics-background-sharing.md; docs/telematics-background-permissions.md; docs/telematics-handover-lifecycle.md; docs/telematics-mobile-dashboard.md; docs/geofence-monitoring.md; docs/fleet-tracker.md; docs/fleet-monitoring-realtime.md.

Research collector: docs/rc-research-data-protocol.md. Current mobile capture implementation: mobile-frontend/services/telematics/rc-research-capture.ts and mobile-frontend/lib/research/rc-research.ts.

Prepared from the repository documentation and current source state on 19 Oct 2026. This guide summarizes current contracts; Laravel Form Requests, services and resources are authoritative for API behavior.

Appendix: Worked Calculations

These examples explain the arithmetic behind the readings and checks in the guide. They are illustrative values, not real vehicle readings or validated safety limits. Rental calibration and RC Research have different storage rules: calibration stores bounded summaries; RC Research preserves consented reviewed samples.

1. Vector magnitude and gravity

A 3-axis vector combines X, Y, and Z. For linear acceleration a = [3, 4, 0] m/s^2:

|a| = sqrt(3^2 + 4^2 + 0^2) = 5 m/s^2

Gravity for each DeviceMotion packet is estimated by subtracting the linear acceleration components from acceleration-including-gravity. Example: including-gravity = [0.30, -9.80, 0.02] and linear acceleration = [0.30, 0.00, 0.02], so:

g = [0.30-0.30, -9.80-0.00, 0.02-0.02] = [0, -9.80, 0] m/s^2

|g| = sqrt(0^2 + (-9.80)^2 + 0^2) = 9.80 m/s^2

The sign and axis depend on phone orientation. This gravity estimate is used as a reference direction, not as a vehicle speed or road slope measurement.

RMS and composite population variation

For vector samples a_i, first calculate each 3D magnitude. The implementation uses equal sample weights:

A_RMS = sqrt(sum(|a_i|^2) / N)

For samples [3,4,0] and [0,0,4] m/s^2, magnitudes are 5 and 4. Therefore A_RMS = sqrt((25+16)/2) = sqrt(20.5) = 4.53 m/s^2.

The composite population deviation removes the mean vector mathematically:

A_STD = sqrt(max(0, A_RMS^2 - |mean(a)|^2))

Here mean(a) = [1.5, 2, 0], whose magnitude squared is 6.25. Thus A_STD = sqrt(20.5 - 6.25) = 3.77 m/s^2. This is population variation over the captured samples, not standard error. Rotation RMS and variation use the same equations in deg/s.

2. Stationary reference, horizontal acceleration, and Forward

A normalized gravity direction d defines the phone-down axis. For any vector v, normalize by dividing by its magnitude:

unit(v) = v / |v|

Stationary first uses the first accepted gravity sample for within-capture projections. After Laravel accepts Stationary, the reusable down reference is d = unit(mean gravity). Forward then learns a horizontal direction relative to d.

Remove the vertical component of acceleration and keep its horizontal component:

vertical_i = a_i dot d; h_i = a_i - vertical_i * d

Example: d=[0,1,0] and a=[1.2,0.5,0]. Then vertical=0.5 m/s^2 and h=[1.2,0,0] m/s^2. Horizontal RMS applies the vector RMS formula to h_i.

Parked-noise Forward threshold

For Forward only, Laravel derives the current minimum as the larger of a fixed acceleration floor and a multiplier times the accepted parked horizontal RMS:

T_forward = max(A_floor, k * H_parked_RMS); current defaults: A_floor=0.5 m/s^2, k=3

Parked horizontal RMS	Calculation	Forward minimum
0.10 m/s^2	max(0.50, 3*0.10)	0.50 m/s^2
0.30 m/s^2	max(0.50, 3*0.30)	0.90 m/s^2
0.36 m/s^2	max(0.50, 3*0.36)	1.08 m/s^2, but Stationary is rejected if other RMS caps fail

The multiplier is an engineering margin, not three-sigma statistics or proof that movement is intentional. The accepted Stationary linear RMS cap remains 0.35 m/s^2. A threshold value does not label a maneuver or establish safe driving.

Forward window mean and coherence

For K contributing horizontal samples h_j:

b = sum(h_j) / K; coherence = min(1, |sum(h_j)| / sum(|h_j|))

Example with h_1=[0.8,0,0] and h_2=[0.6,0.2,0]: sum=[1.4,0.2,0], |sum|=1.414; sum of magnitudes=0.8+0.632=1.432; coherence=1.414/1.432=0.987. The mean b=[0.7,0.1,0] and unit forward direction is approximately [0.990,0.141,0].

Accepted Forward also needs a window of at least 500 ms, at least 5 samples, coherence at least 0.9, and all shared quality checks.

3. Braking, turns, yaw, timing, and coverage

Braking projection

Once forward unit vector f is accepted, longitudinal acceleration is the dot product with f:

longitudinal_i = a_i dot f

Example: f=[1,0,0], a=[-0.65,0.10,0] m/s^2 gives longitudinal=-0.65 m/s^2. The configured braking criterion checks whether the minimum longitudinal value reaches -0.5 m/s^2 or less. No speed check or sustained-braking window is included.

Yaw projection and left/right totals

Yaw rate is the negative dot product of normalized device rotation rate omega and down direction d. Valid intervals integrate with the current rate; intervals over 0.25 seconds are skipped, not filled in:

yawRate_i = -(omega_i dot d); yaw_left += max(0, yawRate_i) * dt; yaw_right += max(0, -yawRate_i) * dt

Example: if d=[0,1,0] and omega=[0,-4,0] deg/s, yawRate=4 deg/s. Over dt=0.25 s, left yaw adds 1 degree. If the rate later reverses, it adds to the right total; opposite motion does not cancel. A left-turn calibration needs at least 20 degrees left and at most 10 degrees right.

Sample rate, coverage, and gaps

The actual mean rate is calculated from accepted sample count and the sensor timestamp span, not the requested interval:

mean_rate_Hz = (N - 1) / ((t_last - t_first) in seconds)

With 101 samples over a 2.0 second span: (101-1)/2.0 = 50 Hz. That is an observed average, not proof the packets were evenly spaced. If elapsed capture time is 2.1 s but sensor span is 1.8 s, coverage=1.8/2.1=85.7%. The largest timestamp/arrival gap is measured separately.

Rental calibration check	Current rule
Sample rate and coverage	At least 8 samples/second and 85% coverage.
Maximum gap	At most 500 ms for shared server quality; Forward continuity breaks over 250 ms.
Gravity and tilt	Mean gravity magnitude 8..12 m/s^2; tilt at most 10 degrees.
Fresh submission	Capture end no more than 60 s old and no more than 10 s in the future.

4. Location, geofence, fleet, and RC calculations

GPS accuracy and geofence confirmation

A fix is evaluated with its reported horizontal accuracy. A location is treated as uncertain near a boundary when it lies within accuracy + 10 m of the circle/polygon edge. Example: 8 m accuracy gives an 18 m uncertainty band. Uncertain fixes keep the last confirmed zone state and reset pending transitions. A membership change needs at least 2 reliable observations spanning at least 10 seconds. First reliable membership is only a baseline.

Location age is computed against server time. Current operational bounds are accuracy <=50 m, age <=60 s, and future skew <=10 s. The authorized fleet display marks a point recent if its timestamp is within 120 seconds; that display threshold is not a GPS update interval. Latest-location retention is 24 hours after the last point; an hourly purge yields a documented maximum of 25 hours when scheduling is running.

RC Research sample rate and feature examples

RC Research requests one update every 20 ms, equivalent to 1/0.020=50 Hz. Actual rate is calculated from timestamps. A summary flags a rate under 8 Hz, a gap over 250 ms, or sensor coverage below 85%. These are data-quality flags, not maneuver labels. A capture needs at least 1 second and 8 samples; buffers stop at 10,000 samples or 4,000,000 JSON bytes.

For a future 2-second window (a pilot candidate only), suppose three acceleration magnitudes are 1, 2, and 3 m/s^2. Their mean magnitude is 2 m/s^2; RMS is sqrt((1+4+9)/3)=sqrt(14/3)=2.16 m/s^2; standard deviation of those magnitudes is sqrt(((1-2)^2+(2-2)^2+(3-2)^2)/3)=0.82 m/s^2. The actual proposed feature set includes axis means/variations, RMS, signed extrema, magnitudes and duration; exact feature definitions must be versioned.

A proposed 1-second stride means windows start one second apart, so adjacent 2-second windows overlap for one second. This helps create examples but does not create independent trials. Train/test splitting must happen by whole session before windows are created to avoid leakage.

What the calculations do not prove

Sensor values, thresholds, geometric projections, GPS accuracy and observer labels describe measurements and quality checks. They do not establish vehicle speed, driver fault, unsafe intent, liability, a validated probability, or a classifier decision. All telematics numeric thresholds remain provisional until physical Android/iOS and controlled trial validation support them. The RC research model is planned, not implemented.

Source guides: docs/telematics-calibration-computation.md, docs/telematics-calibration.md, docs/telematics-device-location.md, docs/geofence-monitoring.md, docs/fleet-tracker.md, and docs/rc-research-data-protocol.md.

Appendix: Consent and Data Collection Rules

Consent is tied to a specific purpose and collection mode. Permission to use a phone sensor or grant OS access is not, by itself, consent to start collection. The server still checks operator eligibility, booking state, release, timestamps and the current consent version.

Who agrees, and when

Flow	Who may operate	Required agreement and start action
Pre-release device setup	Owning Customer for Self Drive; current assigned Driver for With Assigned Driver. Staff first starts handover.	Before checks, the operator sees a setup disclosure. Continue starts the checks. It covers capability/permission checks, one GPS quality fix and manual mount confirmation. These checks do not start tracking.
Rental calibration	Same authorized operator while handover is in progress. Staff supervision in a controlled area is required.	Accept the calibration disclosure for the current operator/rental/profile and consent version. Then manually start each bounded exercise. Motion permission is required; no GPS is requested. Current disclosure agreement is remembered per operator/rental/profile/version.
Released-rental location sharing	Owning Customer (Self Drive) or current assigned Driver (With Assigned Driver).	Explicitly choose “Share in background” or “Share while app is open” and agree to the disclosure. Current location consent policy is version 3. Staff can view latest location; on With Assigned Driver rentals the Customer can also view it.
Prepared start after release	Operator agrees during device setup to a selected sharing mode.	The app holds a one-use instruction in memory. It may start after a fresh server-confirmed release and eligibility check only while the same prepared screen/runtime remains active. Setup itself sends no GPS.
RC Research	Authorized Admin in local/testing RC mode only.	Agree & create session accepts the research disclosure. The Admin then manually taps Start reading and Stop reading. Review and Save trial are separate actions. No rental, booking, GPS or location permission is used.

Choosing Not now, closing a disclosure, denying permission, or not confirming the secure mount does not count as agreement or create a passing result. Background-permission preparation alone never starts collection. Background sharing never silently falls back to foreground mode; the operator can explicitly choose the foreground-only option.

The collection rules

Rule	How the system applies it
Purpose limit	Rental setup/calibration supports readiness before release; active rental GPS provides authorized latest-location visibility and geofence activity; RC Research supports a separate future labeled dataset. Do not repurpose one flow's data for another purpose.
Start only after agreement	Native collection begins only after the explicit action for that flow. Opening a page, login, granted permission or an existing setup pass does not start telemetry. Rental GPS starts only after staff release, except the one prepared foreground start described above.
Least data needed	Setup sends one fresh GPS fix and then discards its coordinates. Rental calibration sends summary statistics, not raw motion arrays. Consented rental location sharing sends latitude, longitude, accuracy and capture time only. RC Research sends reviewed motion samples and annotations, without GPS.
Freshness and server authority	Laravel checks current operator/ownership, booking and handover state, release/receipt state, timestamp freshness, numeric bounds, consent and version on relevant requests. Client screens and phone clocks cannot grant access or decide a server pass.
Foreground/background boundary	Calibration and RC Research are foreground-only. Foreground-only rental sharing pauses when the app leaves foreground. Background rental sharing uses a distinct native task and explicit mode agreement; it is not a cold-start service.
No hidden restart	Leaving the rental screen/runtime, app force-stop, account change, permission/session error or Stop ends/interrupts collection as documented. Return, network restoration, permission grant and app reopen do not restart it automatically. The operator must recover explicitly.
No fabricated readings	Do not fill missing sensor/GPS data with zeros, replay stale fixes, infer an unobserved zone crossing, or turn interruptions into successful evidence. Keep quality and interruption status visible.
No automatic inference of blame	A measurement or geofence event is an observation. It is not proof of speed, unsafe conduct, fault, intent, liability, driver score, penalty or charge. No AI driving classifier is implemented.

Storage, access, stop, and deletion

Data	Storage / who can see it / stop or deletion rule
Setup GPS	Coordinates are evaluated for quality and discarded. Step history stores server outcome, not raw coordinates.
Calibration motion	The app calculates bounded summaries and keeps unsent values in memory only. Laravel stores accepted summary evidence and references; no raw motion stream/table. Cancel, timeout, navigation or backgrounding discards unfinished capture.
Rental latest GPS	One encrypted latest point is saved on the trip with accuracy and timestamp; the session token is in app memory and only its hash is retained. Authorized staff see the latest point; the booking Customer also sees it for With Assigned Driver. Pausing keeps the last point until expiry or receipt/return. It expires 24 hours after the last update; hourly purge makes the upper bound 25 hours if the scheduler is running.
Zone events and alerts	Laravel stores confirmed zone entry/exit event metadata and alert preference, not coordinates or geometry in the event/notification. Default and custom zones are frozen at release. New sharing/resume establishes a fresh membership baseline; crossings during a stop are not inferred.
RC Research samples	Raw reviewed samples are encrypted in private server storage and remain until the owning Admin explicitly deletes them. No age-based purge is implemented. Delete removes the encrypted file and review notes/summaries; a minimal tombstone and session metadata remain to prevent replay. Unsubmitted readings are memory-only and lost when the app closes.
Logs and records	Business/audit logs retain event and consent attribution, mode/version, and server outcomes as needed; do not include coordinates, raw sensor payloads, tokens, identity documents or location history.

Who can collect and access

Laravel is the authorization boundary. The Customer can operate only their eligible Self Drive booking; a Driver can operate only their currently assigned With Assigned Driver booking. Staff starts handover and release. Admin manages geofence defaults and RC Research sessions; Staff can view geofence configurations/activity. Authorized management users with Fleet Monitoring permission can read the map/list. The Customer is a named recipient of latest vehicle location on With Assigned Driver rentals.

If the consent version changes, or the operator stops sharing, the current session cannot continue uploading. A new start must use current terms and a fresh authorized session. Staff receipt clears the latest rental point and invalidates the session. RC Research deletion is explicit and owned by the Admin; the app cannot promise deletion from infrastructure backups.

No contact person or support address is included because the project documentation does not specify one.

Appendix: Data and API Contracts

This section describes the current service boundaries and payload contracts for telematics. Laravel Form Requests, Resources, services, and route definitions are authoritative; frontend Zod schemas mirror client-verifiable rules. API examples below describe shape and field groups rather than a complete copy-paste payload. Standard API responses use the established data envelope; paginated responses also return meta.

1. Mobile setup and calibration contract

Operation	Route / caller	Contract
List eligible rentals	GET /api/mobile/telematics/rentals; Sanctum mobile session, owning Customer or assigned Driver.	Paginated data/meta envelope; default 10, maximum 30 per page. Cards expose booking/vehicle, dates, eligibility and setup progress; omit raw evidence, inspection/signature records, exact GPS and zones.
Current setup	GET /api/mobile/telematics/setup; authorized Customer or Driver.	Returns the most relevant supported booking, or data:null.
Selected booking setup	GET /api/mobile/bookings/{booking}/telematics-setup; booking must belong to Customer or be assigned to Driver.	Returns booking code/state, scheduled timestamps, vehicle, eligibility, setup state, step summaries, calibration projection and trip state. No setup/step IDs, private evidence, exact GPS or zone geometry.
Submit an attempt	POST /api/mobile/bookings/{booking}/telematics-setup/steps/{step}/attempt.	Typed evidence only. Server chooses passed/failed status; top-level status and evidence.status are prohibited. Server enforces operator, confirmed booking, rental timing, staff-started handover, step order, current context and eligibility.
Staff handover/release	Web staff workflow; existing authorized Start Handover and Start Rental actions.	Those management actions control setup start and release. The mobile endpoint cannot release a rental. Start Rental freezes zones and creates the released state transactionally.

Calibration attempt request

For calibration step types, evidence uses protocol_version=3; context is a 64-character lowercase hexadecimal opaque value; consent=true and supervised=true; started_at and ended_at are UTC ISO timestamps with milliseconds. The summary includes forward XYZ/duration/count/coherence, sample_count, sensor_duration_ms, max_gap_ms, linear RMS/mean/STD, horizontal RMS, rotation mean/RMS/STD/peak, longitudinal min/max, left/right yaw degrees, maximum tilt, and mean gravity XYZ. Raw sample arrays and extra keys are rejected. Laravel computes the result and appends the attempt; retry history is preserved.

2. Rental location monitoring contract

All mobile paths below use Sanctum bearer authentication and mobile throttling. Laravel rechecks booking ownership/operator, current release and trip state, consent version, session token, fix freshness and accuracy on every relevant mutation.

Method and path	Request contract	Response / effect
GET /api/mobile/bookings/{bookingCode}/location-monitoring	No coordinate or token input.	Minimal state: bookingCode, canMonitor, tripStatus, lastLocationAt, server policy. No coordinates or token. Cache-Control: no-store.
POST .../location-monitoring/start	latitude [-90,90], longitude [-180,180], accuracy [0, configured max], captured_at; consent accepted; current consent_version=3; sharing_mode is foreground or foreground_background. session_token is prohibited.	Returns the minimal resource plus sessionToken once. Rotates a 64-character ephemeral token; server stores only its hash. Start reuses the one trip if it already exists.
POST .../location-monitoring/location	Same GPS fix fields plus current 64-character session_token. speed, heading, samples and current_zone are prohibited.	Accepts a fresh latest point; encrypted latest-location value and lastLocationAt update. Duplicate/older captured seconds cannot replace latest state or change geofence membership.
POST .../location-monitoring/pause	Current session_token; optional reason is manual or interrupted.	Pauses sharing without ending booking/trip. The final encrypted latest point stays until it expires or the vehicle is received/returned.

The started flow uploads no route, raw sensor stream, speed, heading or client-claimed geofence event. An accepted point contains latitude, longitude, accuracy and captured_at. Native dates use ISO 8601. Actual upload rate is capped at one per five seconds; fix age is at most 60 seconds with at most 10 seconds future skew and accuracy at most 50 m under current configuration.

Driver contract alignment note

The documented system contract supports the owning Customer on Self Drive and the current assigned Driver on With Assigned Driver. In the current source, LatestLocationRequest::authorize() checks for the Customer role, and the start/pause requests inherit that check. That appears narrower than the documented Driver contract. Reconcile this guard and its tests before relying on Driver location submissions; this PDF records the discrepancy rather than claiming it is resolved.

Geofence contract

Client payloads cannot claim a zone or event. Laravel evaluates each frozen circle/polygon from accepted fixes; the first reliable fix establishes a baseline. Confirmed transitions write a server-owned geofence_enter/geofence_exit event and may create a named in-app alert. No coordinates or geometry are copied into the event or notification.

3. Web Fleet Monitoring and event contract

Operation	Contract
GET /admin/fleet-tracker	Stateful web session; active Admin/Staff with view fleet tracker permission; no-store. Query: search (optional, max 160 characters), status, rental_type (self_drive or owners_driver), page (>=1), per_page (1..100, default 15). Filtering occurs before pagination.
Response	data array plus meta: current_page, last_page, per_page, total filtered records, total_ongoing before filters. Each item exposes booking id/code, customer name, rental type, start/end, vehicle (id/name/plateNumber or null), and tracking (status/tripStatus/lastLocationAt/location).
Map location	Nullable {latitude, longitude, accuracy, expiresAt}; only authorized, released, ongoing, consented, unexpired latest points are returned. No route, sensor, token, consent record, financial data, documents or inspection evidence. Map represents the current filtered page (default 15).
Realtime notice	After commit, private-fleet-monitoring broadcasts empty fleet.changed notices. The event contains no booking id, coordinates, customer info or token. Web client uses notice only to trigger a new authorized HTTP read.
Channel authorization	POST /admin/broadcasting/auth uses web cookie/CSRF, active Admin/Staff and the same fleet permission. Exact private channel and bounded socket id are validated. A mobile bearer token cannot authorize the web channel.

Tracking state is a saved-record projection, not proof of live connection. The recent_location cutoff is 120 seconds. Browser reconciles a visible tab every 30 seconds; Reverb notices coalesce for 750 ms; subscription errors retry after 10 seconds; reads time out after 10 seconds. These are web refresh behaviors, not GPS sampling settings.

4. RC Research API contract

Operation	Route	Key contract
Access/config	GET /api/mobile/rc-research	Restricted research token, active Admin, local/testing environment and RC profile. Returns server protocol/settings; protocol version, bounds and consent version are server-owned.
Create session	POST /api/mobile/rc-research/sessions	Name, surface, slope, mount orientation, battery condition, platform, app_version and consent=true. Server owns owner, ID, added_load_g=0 and protocol. Client cannot submit user_id, protocol or added_load_g.
Save trial	POST /api/mobile/rc-research/sessions/{session}/trials	Multipart metadata plus samples_json. Trial id, protocol_version=1, intended/observed movement and intensity, label_status, quality, event interval, notes, and UTC start/end. summary, samples_path and user_id are prohibited.
List trials	GET /api/mobile/rc-research/trials?page=...&per;_page=...	Paginated owner-bound metadata and summaries only; per_page max 30. Raw samples, filesystem path, checksum and owner identifier are not exposed.
Delete trial	DELETE /api/mobile/rc-research/trials/{trial}	Owning Admin explicit delete. Encrypted sample file removed; minimal tombstone remains to block replay.

Sample contract

samples_json is a bounded JSON list (8..10,000 samples; at most 4,000,000 characters/bytes per current request limit). Each sample has exactly: elapsed_ms; acceleration_timestamp_s; gravity_timestamp_s; rotation_timestamp_s; linear_x/y/z; including_gravity_x/y/z; rotation_x/y/z; orientation. Values must be finite and within field bounds; orientation is one of -90, 0, 90, 180. Missing channels are not filled with zeros. Server validates sample order/timing, movement-intensity pairing, annotation interval and ownership before writing an encrypted immutable private file.

5. Persisted data boundaries and error behavior

Data store / record	What belongs there	What does not belong there
telematics_setup_sessions	One pre-release setup lifecycle per supported booking; status, timing and release state.	Raw sensor streams, customer location history or editable zone geometry.
telematics_setup_steps	Append-only typed attempt history and bounded calibration evidence; current operator/latest attempts determine readiness.	Raw motion arrays, client-selected outcomes, exact setup GPS.
telematics_trips	At most one post-release trip. Current consented location flow stores one encrypted latest point, consent/mode metadata, token hash and latest timestamp.	A route in latest_location, raw motion stream, plaintext duplicate of the encrypted point, or session token.
telematics_location_points	Route history for the separate legacy/history ingestion contract where applicable.	The consented latest-only location flow does not append to this table.
telematics_events + geofence state	Significant lifecycle/confirmed zone observations and per-zone transition state.	Continuous raw GPS or precise coordinate copies in notifications.
RC Research private files + metadata	Separate Admin-owned reviewed raw samples (encrypted file) and validated trial/session metadata.	Rental trip data or samples in database JSON/public storage.

Validation and response rules

Laravel Form Requests are authoritative. Zod provides immediate client feedback but does not replace server validation. Untrusted extra keys and client-selected pass/fail or current_zone are rejected where the contract prohibits them. Field failures return validation errors; the client should map Laravel 422 fields to the form. Authorization, authentication, conflict and not-found status meanings remain distinct. Idempotent RC trial retries require the same ID and same content; changed content or a deleted ID conflicts. Setup attempts are append-only; failed evaluated captures are real attempts, while native/transport interruption does not fabricate an outcome.

These contracts summarize the repository on 19 Oct 2026. Read the relevant Laravel route, Form Request, Resource and service before changing a payload. The Driver location-request authorization mismatch above is a source/documentation gap requiring implementation reconciliation.

Appendix: Sensor Functions and Calculations

The phone does not hand the app a vehicle-ready “speed” or “tilt” reading. The app receives platform-processed motion channels, checks their timestamps, converts them into a phone-device coordinate frame, and derives bounded summaries. These measurements support setup calibration or RC research; they do not prove driving behavior.

What each sensor channel does

Channel	What it measures	How JamJade uses it
Linear acceleration (accelerometer)	DeviceMotion acceleration with gravity removed; XYZ in m/s^2. This represents non-gravitational acceleration in the phone frame.	Computes 3D magnitude, linear RMS/mean/variation, horizontal acceleration and projected longitudinal acceleration for the calibration exercises. It is not vehicle speed.
Acceleration including gravity	Platform acceleration vector with gravity included; XYZ in m/s^2.	The app subtracts linear acceleration component by component to estimate gravity. This helps find the phone-down direction and measure tilt relative to a reference.
Rotation rate (gyroscope)	Angular rate around device axes; normalized to XYZ in degrees/second. DeviceMotion provides alpha/beta/gamma labels, which are mapped differently by iOS and Android.	Projects rotation onto the gravity direction to estimate short-window yaw-left and yaw-right totals. The axis/sign mapping still requires physical validation on both platforms.
Screen orientation	A screen-layout orientation value supplied with a sample.	Stored with RC Research samples as context. It is not the phone’s full 3D attitude, vehicle heading or a substitute for gravity reference.
Magnetometer / GPS speed	Not used by this motion-calibration calculation.	No magnetic compass heading or speed is calculated from accelerometer/gyroscope readings. RC Research does not request location.

All three DeviceMotion channel timestamps must be finite and within 100 ms of one another for a calibration packet. The app does not assume alpha, beta and gamma are portable XYZ labels. In the inspected adapter, iOS maps rotation rate to [gamma,beta,alpha]; Android maps it to [alpha,beta,gamma].

Gravity direction and tilt

For a sample, estimate gravity by subtracting the acceleration vectors, then normalize it to a unit direction:

g_i = accelerationIncludingGravity_i - linearAcceleration_i; d_i = g_i / |g_i|

The first accepted stationary sample supplies a temporary direction during that capture. After Laravel accepts Stationary, the saved down reference is d = mean(g) / |mean(g)|. For each later sample, compare its normalized gravity direction d_i to the saved direction d:

tilt_i = acos(clamp(d_i dot d, -1, 1)) * 180 / pi; max_tilt = max(tilt_i)

Example: if the dot product between current and reference directions is 0.9962, tilt = acos(0.9962) * 180/pi, approximately 5 degrees. Calibration allows at most 10 degrees under the current standard profile. Tilt is a change in gravity direction, not absolute phone attitude; a flat phone can rotate in its horizontal plane without gravity alone detecting that turn.

How readings become calibration results

Each exercise uses the appropriate projection and summary. Laravel receives bounded evidence and decides whether the exercise passed. Raw motion sample arrays are not submitted by rental calibration.

Exercise / value	Calculation or role	Current standard check
Stationary reference	Mean gravity direction supplies down (d). Full-interval acceleration and rotation statistics form the parked-noise baseline.	6-second capture; linear RMS <=0.35 m/s^2; rotation peak <=6 deg/s; left+right yaw <=5 degrees.
Vertical and horizontal acceleration	vertical_i = a_i dot d; h_i = a_i - vertical_i*d. Horizontal RMS summarizes acceleration perpendicular to gravity.	Supports the Forward learner and the accepted parked horizontal-noise baseline.
Forward direction	Find a consistent qualifying horizontal-acceleration window. b = mean(h); f=b/|b| after server acceptance. coherence=min(1, |sum(h)|/sum(|h|)).	Threshold=max(0.5 m/s^2, 3*parked horizontal RMS); window >=500 ms, >=5 samples, coherence >=0.9.
Braking	longitudinal_i = a_i dot f. Track the minimum signed longitudinal value.	Minimum <= -0.5 m/s^2 in the expected direction.
Left/right turns	yawRate_i = -(rotationRate_i dot d). Integrate positive and negative portions separately using sensor timestamp deltas.	Expected side >=20 degrees; opposite side <=10 degrees. Integration skips gaps over 250 ms.
Summary quality	Count/span/rate/coverage/gaps, gravity magnitude, mean vectors, RMS, peak and max tilt validate the bounded capture summary.	At least 8 samples/s; at least 85% coverage; max gap <=500 ms; gravity magnitude 8..12 m/s^2; tilt <=10 degrees.

A yaw example

Let down d=[0,1,0] and normalized gyroscope rate omega=[0,-4,0] deg/s. Then yawRate=-(omega dot d)=4 deg/s. For a valid 0.25-second interval, yaw-left increases by 4*0.25=1 degree. If the phone turns the opposite way later, yaw-right increases separately; left and right totals do not cancel. An interval longer than 0.25 seconds is skipped rather than filling missing data.

Different capture paths store different data

Rental setup calibration requests DeviceMotion at 50 ms (20 Hz), but stores summary evidence only: statistics, references and evaluated result. It does not retain raw arrays. RC Research requests updates every 20 ms (50 Hz) and, after separate Admin consent, review and explicit save, retains sample-level motion values in an encrypted private file. Its readings are still platform-processed, and actual sample rate may differ from the request.

Key outputs and what they mean

Output	Plain meaning
RMS	Typical 3D signal magnitude including sustained mean; acceleration uses m/s^2, rotation uses deg/s.
STD	Composite population variation around the mean vector; not a standard error or a custom filter.
Mean vector	Signed average on each phone-device axis. It does not automatically correct sensor bias.
Maximum gap / coverage / mean rate	Evidence about timestamp span and missing readings. Requested Hz is not guaranteed delivery.
Coherence	Directional agreement within a Forward window, from 0 to 1; not probability/confidence.
Classifier output	None exists in the current system. These calculations are not an AI driving classification or a finding of fault.

Reference: docs/telematics-calibration-computation.md and docs/rc-research-data-protocol.md. Limits are provisional and require physical Android/iOS validation.

Emergency Contact and Self Drive Eligibility

An emergency contact may be skipped during onboarding without making the customer ineligible for Self Drive.

Self Drive eligibility

Skipping the emergency-contact step does not block Self Drive eligibility or booking. The skip is recorded separately on the customer profile; it is not part of the Self Drive document-approval check.

Requirement	Current rule
Minimum age	At least 18 years old.
Required documents	Driver's license and proof of address, both approved by staff.
Emergency contact	Optional for Self Drive. The customer can save one contact or skip for now.

Emergency-contact data contract

If the customer supplies a contact, the account stores one contact with a name, relationship, and phone number. The phone number must contain 11 digits and start with 09. Saving a contact clears the profile’s skipped marker. If the customer skips, no contact details are submitted in that step; the profile records that it was skipped.

This note reflects the current application rules. Rental release still follows the separate staff-controlled handover and Start Rental process described in this guide.